

Tree-Based Classifier for Hyperspectral Image Classification via Hybrid Technique of Feature Reduction

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Abstract—Remote sensing by hyperspectral imaging (HSI) has proved to be an emerging technology for researchers with broad appeal in legionary branches. Imaging spectroscopy, another name of hyperspectral remote sensing, is mainly employed to identify ground cover. HSI contains eminent spectral details, leading to a high dimension which results in a menace for researchers. A viable resolution to this crux is feature reduction (FR). Hither, a hybrid technique of FR, mainly coupling of feature extraction (FE) with feature selection (FS) has been preferred. First, features are extracted via Principal Component Analysis (PCA), then selected via Mutual Information (MI). At the last step, classification has been carried out using tree-based classifier. Here decision tree (DT), random forest (RF) and extra tree (ET) have been chosen as the classifier. For this study, AVIRIS which is also known as the Indian Pine dataset has been picked. This proposed approach offers classification accuracy(CAR) of 93.48% for DT, 96.83% for RF and 97.93% for ET.

Keywords—Hyperspectral imaging, principal component analysis, mutual information, decision tree, random forest, extra tree.

I. INTRODUCTION

Hyperspectral Imaging has emerged to be a prevalent remote sensing technology to the researchers. NASA's Airborne Visible Infrared Imaging Spectrometer (AVIRIS) gathers information utilizing images via band of spectrum. Afterward, the captured images are assembled for the formation of a hyperspectral cube. AVIRIS constitutes of 224 bands with a spectral resolution of 0.1 μm [1]. This narrowness of HSI implies high-dimension data with numerous research scopes. But these contiguous narrow bands introduce an offshoot, known as Hughes phenomenon [2]. The auspicious measure to ease this offshoot is FR which is also coined as dimension reduction(DR). Two basic deftness constitute FR and they are FE and FS [3][4]. FE is the transformation contrivance that sorts out features via mapping founded on criterion like variance [5][4]. On the contrary, FS is the procedure of selecting the highest informative feature sets. Principal Component Analysis (PCA), a linear FE deftness has been picked. Because this performs well for the strongly adjoining

dataset as AVIRIS. But the primary pullback of PCA is that the class separability is not taken into account while performing eigen analysis (EA). Therefore, it is not certain that the top features have been chosen for the separation of classes[6]. Another drawback of PCA is that small objects which do not contribute in a significant way to total variance are not taken [7]. Therefore, if FS is coupled with FE, then higher CAR can be achieved. Therefore, in this outlet, the coupling of PCA and Mutual Information (MI) have been recherche so that it gives better CAR with three tree-based classifiers.

The research paper of us is impersonated in the following way. **Section II** has to do with the related works. Our approach and methodology are illustrated in **Section III**. **Section IV** contains the research process along with experimental result analysis. In fine, the research is wrapped up in **Section V**.

II. RELATED WORKS

A sensor is defined on the narrowness and the closeness nature of mensuration [4]. Due to this, HSI shackled with both chances and penalties. One of the grave penalties is Hughes Phenomenon [2]. The second fancy name of this penalty is 'curse of dimensionality', coined in [8]. Here, Richard Bellman defended the argument that the curse befell while forming test data that do no emerge in lower dimension but appear in higher dimension. FR is the viable resolution that comes handy to lessen this crux [3]. In [9], PCA scheme and in [10], various MI measures have been practised. [7] is a very early on HSI classification work where PCA and MI were used in a collaborated manne. They highlighted that some crucial information may be lost if PCA is solely employed. On the other hand, CAR was not significant while utilizing MI only. Therefore, PCA was followed by MI for FR. They achieved a CAR of 80% considering 10 features of AVIRIS. In [3], Supervised Non-parametric MI, supervised Parametric Bhattacharya distance, Unsupervised Signal Noise Ratio were used for selecting features and for FE Decision

Boundaries, Linear Discriminant Analysis (LDA), Regularized LDA, Nonparametric Discriminant Analysis and PCA were recherche. PCA was employed as a preprocessing step and then FS was guided by nMI in [11]. By doing so, they obtained 96% CAR for HYDICE dataset but only for 5 features for Support Vector Machine (SVM) classifier. In [12], similar tasks were found but only with the AVIRIS data. The paper attained eminent CAR using SVM for limited number of features. In [13], a hybrid technique was achieved by coupling PCA with nMI and qMI . They finally got 87.68% CAR for AVIRIS data containing 14 classes by applying SVM. The hybrid technique with PCA and nMI were followed in [14] with SVM classifier, and they got 97.2662% CAR for AVIRIS and 99.2074% CAR for HYDICE data. A very early comparison based work among Artificial Neural Network, Maximum Likelihood, Spectral Angle Mapper and DT on AISA data was evaluated in [15]. In [16], DT with cross-validation (CV) was performed on AVIRIS Data along with other classifiers. Paper [17] showed a state of the art ensemble technique, rotation random forest with kernel PCA. Here they implemented linear, radial basis function (RBF) and polynomial kernels for kPCA and the best result was found using the polynomial kernel. In [18], a combination of two enhancement of RF was cascaded and compared it with other RF and SVM. They found that Cascaded RF provided better experimental results.

III. METHODOLOGY

The structural outline of the methodology of this research is exhibited in figure 1.

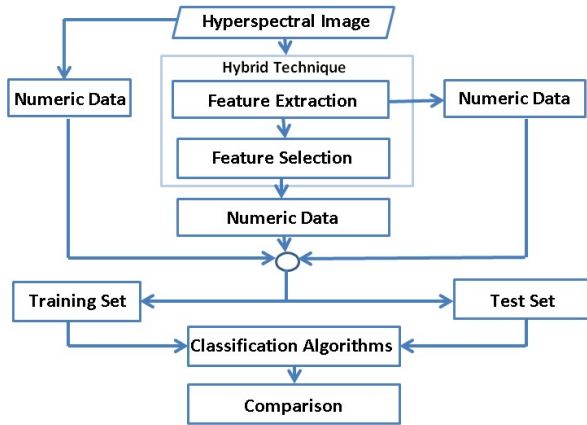


Fig. 1. Flow diagram of our research methodology

A. Principal Component Analysis

FE is one of the two procedure of FR and PCA is a tremendously famed FE technique. PCA attenuates the dimension of highly tied in features into untied orthogonal feature set via transformation functions [9]. These are exlaimed as Principal Components (PCs). Any feature is interpreted as:

$$A = E^T B \quad (1)$$

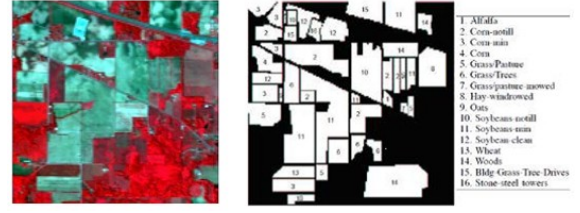


Fig. 2. RGB (left) and Ground Truth (right) image of AVIRIS

Here, B , A and E^T signify input image, mapped image and the transpose of normalized eigen vectors respectively. The eigen vectors are itemized in descending order based on eigen values. This implies that the maximum eigenvalue is in first PC. Only a limited number of PCs contribute to total variance and that's why only these PCs are taken.

The evaluation procedure to find principal components are impersonated in the following way. The very first step is to deduct the mean from each dimension after getting the dataset and then the average is calculated in such a manner that the norm of the dataset becomes zero. The newly generated data is labelled data adjust. The enumeration of the covariance matrix is the succeeding step. After this, the eigen values and eigen vector are quantified and sorted in descending order. This is the proceeding which extracts the lines that gives handy details of the data. In dimension reduction step, the feature vector is constructed by keeping the eigen vectors which have superior eigen values. In the last stride, the transposed feature vector is multiplied with transposed data adjust to spawn the final data.

B. Mutual Information

The function that numbers the dubiety of a random variable is termed entropy. If A and B are two variables and the probability of A and B are $P(A)$ and $P(B)$ respectively. The entropy of A i.e $S(A)$ is interpreted as:

$$S(A) = - \sum_A P(A) \log P(A) \quad (2)$$

The joint entropy of A and B i.e $S(A, B)$ is interpreted as:

$$S(A, B) = - \sum_A \sum_B P(A, B) \log P(A, B) \quad (3)$$

Here, $P(A, B)$ refers to the joint probability of A and B . MI quantifies the reliance between two variables and enumerates in which content one variable perceives another. Then the MI [19] between A and B is

$$MI(A, B) = S(A) + S(B) - S(A, B) \quad (4)$$

The MI between two images P and Q is interpreted as:

$$MI(P, Q) = - \sum_{N1} \sum_{N2} P(p, q) \log (P(p, q) / P(p)P(q)) \quad (5)$$

Here $p \in P$ and $q \in Q$ and $N1, N2$ are the largest pixel values of P and Q in order. Both the joint entropy and marginal entropy climbs up exponentially with the escalation of dimension. [20].



Fig. 3. PC1(left) is the brightest and the hybrid technique choses PC4(middle) over PC2(right)

TABLE I
TRAIN DATA AND TEST DATA

CLASSES	TRAIN	TEST
Corn	23	6
Grass/Pasture	76	19
Soybean-notill	104	26
Soybean-clean	64	16
Bldg-Grass-Tree-Drives	61	15
Hay-windrowed	82	21
Corn-notill	88	22
Grass/Trees	48	12
Soybean-min	105	26
Corn-min	58	14
Woods	100	25

TABLE II
TOP TEN FEATURES PICKED VIA HYBRID TECHNIQUE

Listed Method	1st	2nd	3rd	4th	5th
Hybrid Technique	PC1	PC3	PC4	PC2	PC5
Listed Method	6th	7th	8th	9th	10th
Hybrid Technique	PC6	PC9	PC7	PC15	PC14

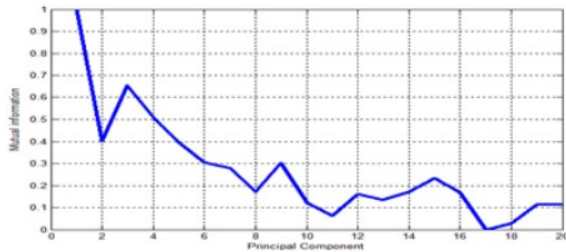


Fig. 4. MI between class label and first 20 PCs

C. Decision Tree

DT is a supervised Machine Learning (ML) procedure that sets up a model for regression and classification by splitting the data into subsets and illustrates those subsets in the anatomy of a tree. Here, the class is foretold based on decision rules taken from input data. The root indicates the best feature by calculating attribute selection meters like Information gain or Gini index. It can operate on numerical data as well as categorical data. Moreover, outliers and missing values barely have any weight on the model's data. But DT works on greedy

approach which occasionally guides to overfitting.

D. Random Forest

RF is a tremendously renowned ensemble ML algorithm that buds from DT. It can control the overfitting offshoot of DT while building a model. Numerous classification models are produced where every model is assembled via a feature selector like Gini Index, Information Gain and Gain Ratio. These models discretely learn and contribute to the prediction. The ultimate result is molded from predictions that were gained from learning by the models.

E. Extra Tree

Extremely Randomized Trees or ET is another ensemble learning operating procedure such as RF. ET merges the output of a significant number of uncorrelated trees to yield CAR. The first of the two fundamental distinctions between ET and RF is that ET samples barring replacement. The second one is that it goes for the random features to split the tree nodes over the best one. Moreover, ET is superior to RF in the context that ET is quicker and barely allows noisy data.

F. Procedure of Hybrid Technique

It is above said that the hybrid approach puts in FS on extracted features to attain finer CAR. This section gives a brief narration of the procedure of hybrid technique.

- At first, features are extracted from the input image via *PCA*
- After this, class labeled data is created from extracted features.
- Then *MI* is quantified from the class labeled data.
- Next, features are sorted in descending order on the basis of value of *MI*.
- Finally, classification is performed using *DT*, *RF* and *ET*.

Experimental analysis are revealed in the upcoming section.

IV. EXPERIMENTS

A. Experimental Data

All the procedures for this research has been appraised with AVIRIS 92AV3C data, seized by a hyperspectral sensor. This comprises of 220 band with spectral resolution of

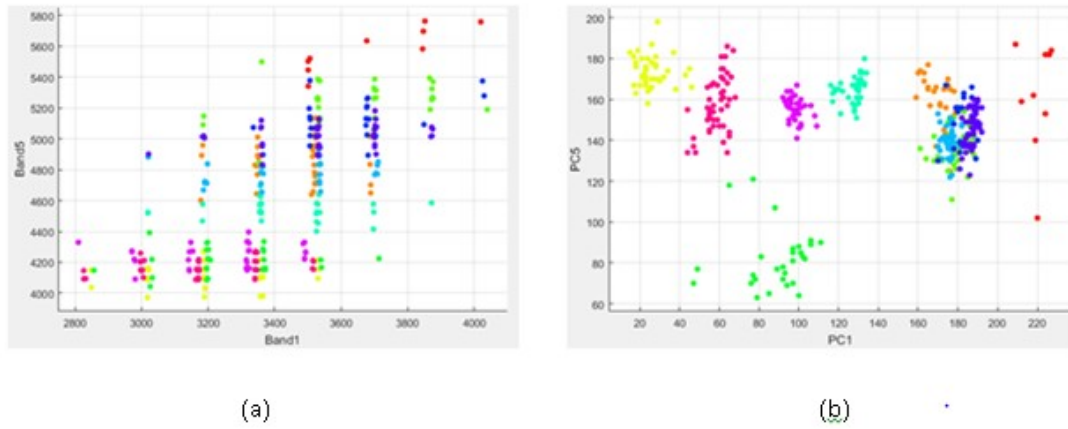


Fig. 5. Projection of Band1 and Band5 in (a) before and PC1 and PC5 in (b) after applying Hybrid Technique

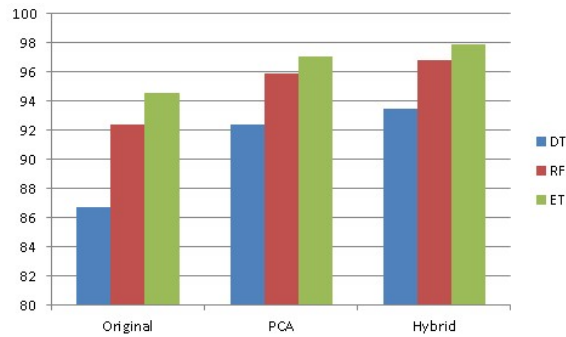


Fig. 6. Classification Accuracy

0.4 μ m-2.5 μ m containing 16 classes. On account of sparse sample, alfalfa, grass/pasture-mowed, wheat, oats, and stone-steel towers classes were not considered. Dataset interpretation are exhibited in Table I. These samples were taken on the basis of ground truth image of AVIRIS. Figure 2 exhibits the Ground Truth Image and RGB Image of AVIRIS. Numeric values for classification were generated from AVIRIS using MutliSpec software developed by Purdue University [21].

B. Hybrid Technique Results

In our work, dimension of AVIRIS is lowered to 20 after applying *PCA*. Figure 3 exposes that PC1 is the most vivid one as it has maximum variance. However, PC4 is tabulated greater than PC2 as it provides superior information. Figure 4 illustrates the *MI* of first 20 PCs. The top ten selected features of AVIRIS after going through hybrid technique is sorted in Table II. The ease of using hybrid approach is perceptible in Figure 5. It is simpler to split the classes in (b) than (a) in Figure 5.

C. Classification Results

AVIRIS sensor captured the data from agricultural parts of Indian Pine. Therefore, the classes mainly constitutes of different kind of vegetation. In our research, three classifiers have been employed to classify eleven classes which

are Corn, Grass/Pasture, Soybean-notil, Soybean-clean, Bldg-Grass-Tree-Drives, Hay-windrowed, Corn-notill, Grass/Trees, Soybean-min, Corn-min and Woods. For the purpose of classification, Google Colaboratory [22] was utilized which is a cloud-based service delivered by Google. To remove the overfitting or maybe the underfitting problem and to evaluate the model correctly, 10-fold cross-validation [23] technique was employed. We partitioned our entire dataset into an 80:20 proportion. After that, we exercised tree-based classifiers such as DT, RF and ET where Gini impurity was nominated in each classifier and 100 trees were chosen for RF and ET. For original data, DT, RF and ET yielded 86.74%, 92.37% and 94.56% accuracy respectively. Other evaluation metrics like roc-auc, f1-score, precision and recall are depicted in Table III. PCA was applied to lessen the dimension of the features. After using PCA, the classification accuracy increased to 92.43%, 95.88% and 97.08% in DT, RF and ET in order. Other evaluation metrics are depicted in Table IV. Then, MI was adopted for further improving the performance. It is perceptible from Table V that the overall performance has indeed raised via hybrid technique. The graphical representation of CA of all the approaches is exhibited in Figure 6. The ET classifier outperformed other classifiers with an accuracy of 97.93% which is highlighted in Table V.

TABLE III
CLASSIFICATION RESULTS OF ORIGINAL DATA

Classifier	Accuracy	Precision	Recall	F1 score	ROC-AUC
DT	86.74%	88.19%	86.89%	87.08%	92.10%
RF	92.37%	94.35%	93.95%	93.39%	99.68%
ET	94.56%	95.75%	95.53%	95.63%	99.84%

TABLE IV
CLASSIFICATION RESULTS AFTER APPLYING PCA

Classifier	Accuracy	Precision	Recall	F1 score	ROC-AUC
DT	92.43%	93.61%	94.06%	91.69%	95.55%
RF	95.88%	96.56%	97.11%	96.59%	99.91%
ET	97.08%	97.54%	97.41%	97.07%	99.94%

TABLE V
CLASSIFICATION RESULTS AFTER APPLYING HYBRID
TECHNIQUE

Classifier	Accuracy	Precision	Recall	F1 score	ROC-AUC
DT	93.48%	94.57%	94.45%	93.40%	96.58%
RF	96.83%	97.15%	97.21%	96.85%	99.93%
ET	97.93%	98.16%	98.01%	98.29%	99.96%

V. CONCLUSION

A Hybrid approach that is the coupling of FS and FE leads to effective subspace detection that gives better classification. Three tree-based classifiers, i.e DT, RF and ET, were implemented to classify the data. In every approach mentioned in this paper, ET outperformed both DT and RF. ET gave the highest accuracy of 97.93% via hybrid technique. One limitation of our work is that in hybrid technique, *PCA* was coupled with MI. But, *PCA* only works on linear data set and hence may not succeed to attain better accuracy. Here, Kernel PCA (*KPCA*) can be the way to overcome this offshoot. Moreover, further, improvement can be introduced by exploring other MI measures.

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